Zihan Hei

Statistics Undergraduate Student

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Education

AUG 2023 – JUN 2027

Bachelor of Arts in Statistics

Binghamton University, State University of New York

Minor in Digital and Data Studies | GPA: **3.8**/4.0

Research Experience

NOV 2025 - PRESENT

Transcriptomic Biomarker Discovery and Validation in Chemo-Immunotherapy

Independent Study with Dr. Nancy Guo | Binghamton University

- Reproduce and validate RNA-seq and single-cell RNA-seq preprocessing workflows, including quality control, normalization, gene filtering, and highly variable gene selection.
- Apply NMF, differential expression analysis, cross-study comparisons, and GSEA to identify and validate treatment-response biomarkers and biological pathways (manuscript in preparation).

AUG 2025 - PRESENT

Machine Learning for Health Information Avoidance

Independent Study with Dr. Heather Orom, Dr. Shane McCarty, and Dr. Kargin Vladislav | Binghamton University

- Clean, recode, and score sociodemographic, psychological, health behavior, and media-use variables in R.
- Develop and compare regression, random forest, and MARS models using cross-validation and predictive performance metrics; published a reproducible Quarto report online (manuscript in preparation).

SEP 2025 - APR 2026

A Mixed-Methods Analysis of Promotion and Prevention Mindsets

Undergraduate Research with Dr. Shane McCarty and 3 peers | Binghamton University

- Applied logistic regression modelling in R to investigate whether promotion and prevention mindsets predict health and safety-related goal preferences.
- Collaborated on a mixed-methods research project combining quantitative analyses with qualitative coding to examine public conceptions of community health and safety.
- Contributed to a 2025 SCRA presentation and served as first author on a poster abstract accepted for the 2026 APHA Annual Meeting.

AUG 2023 - DEC 2024

First-Year Research Immersion: Community and Global Public Health

Undergraduate Research | Binghamton University

- Collaborated with a seven-person team on women's health, mental health, vaccination, and reproductive health research.

- Conducted SPSS analyses, designed and distributed a Qualtrics survey, and presented findings through two FRI poster presentations.

JUN 2025 - PRESENT

Political Affiliation and Zero-Sum Beliefs

Undergraduate Research with Dr. Shane McCarty and 2 peers | Binghamton University

- Examine political party affiliation and zero-sum beliefs as predictors of voter choice in the 2024 U.S. election.
- Conduct ANOVA/Kruskal-Wallis tests, logistic regression, and machine-learning classification in R (manuscript in preparation).

Professional Experience

JUN 2026 - PRESENT

Quantitative Intern, First-Year Research Immersion Program

With Dr. Megan Fegley and Dr. Caitlin Light | Binghamton University

- Contribute to R Shiny dashboards for student outcomes and research dissemination metrics used in program evaluation and reporting.
- Develop an integrated dissemination database and Qualtrics surveys to collect publication and presentation outcomes across FRI programs.

JUN 2026 - PRESENT

Mental Health Workshop Quantitative Intern

Summer Internship with Dr. Shane McCarty | Binghamton University

- Designed Qualtrics surveys and quantitative evaluation methods assessing mental health literacy, well-being, nonviolent communication, and workshop effectiveness.
- Created interactive data visualizations and individualized wellness profiles using Observable JS.
- Developed an interactive wellness webpage integrating workshop materials, participant profiles, and trainer-facing visualizations for program evaluation.

Teaching & Mentoring Experience

JAN 2025 - MAY 2025

R Programming Peer Mentor

First-Year Research Immersion Program | Binghamton University

- Mentored undergraduate students in foundational R programming, data analysis, code troubleshooting.
- Collaborated with the instructor and other research assistants to support research teams' Quarto research reports

Publications

1. McCarty, S., Light, C., **Hei, Z.**, Panzik, J., Wolfe, P., & Fegley, M. (2026). *FRI Semester 3 Peer Mentor Mindsets Guide*. Binghamton University. shanemccarty.github.io/FRImindsets3/

2. McCarty, S., Light, C., **Hei, Z.**, & Fegley, M. (2026). *FRI Peer Mentor Mindsets Guide.* Binghamton University. shanemccarty.github.io/FRImindsets

3. McCarty, S., & **Hei, Z.** (2025).
Chapter contributions to *The Quantitative Playbook for Public Health Research in R*, S. McCarty (Ed.).
  - McCarty, S., & **Hei, Z.** "Reproducible Reports."
  - **Hei, Z.**, & McCarty, S. "Import Data Once."
  - **Hei, Z.**, & McCarty, S. "Read Me."
  - **Hei, Z.** "Writing Methods & Results Reproducibly."
  - **Hei, Z.** "APA Citations & References."
  - **Hei, Z.** "Publish Your Report to Quarto Pub."

shanemccarty.github.io/FRIplaybook/

Presentations & Conference Abstracts

2026

A Mixed-Methods Analysis of Promotion and Prevention Mindsets

American Public Health Association Annual (APHA) Meeting

- Hei, Z., Giallella, J., & McCarty, S. First-author poster abstract accepted for presentation. https://apha.confex.com/apha/2026/meetingapp.cgi/Paper/594471

NOV 2025

Promotion & Prevention Mindsets: A Four-Study, Mixed-Methods Approach

Society for Community Research & Action (online)

- Invited presentation; McCarty, S., Hei, Z., Rualo, G., Khalap, I., & Giallella, J.

DEC 2024

Public Knowledge & Attitudes on Vaccinations and Reproductive Health

FRI Proposal Poster Research Session 11th annual, Binghamton University

DEC 2023

Endometriosis and Women's Mental Health

FRI Proposal Poster Research Session 10th annual, Binghamton University

Data Science Projects

JAN 2026 - MAY 2026

Histopathology Thyroid Cancer Detection Using YOLO26 | Group Project

Digital & Data Studies Group Project | Binghamton University

- Built a YOLO26-based deep-learning model to classify thyroid histopathology images and evaluated confusion matrices, Type I/II errors, class imbalance, and dataset representativeness.

SEP 2025 - DEC 2025

Deep Reinforcement Learning in Board Games | Group Project

Digital & Data Studies Group Project | Binghamton University

- Implemented PyTorch reinforcement-learning agents for Tic-Tac-Toe and Connect Four through self-play and visualized training performance.

Awards, Honors & Scholarships

FEB 2026 - FEB 2027

NSF Research Experiences for Undergraduates (REU) Research Stipend Recipient

Binghamton University

- Year-long research stipend supporting transcriptomic biomarker discovery and validation in chemo-immunotherapy with Dr. Nancy Guo.

SUMMER 2026

Fleishman Internship Funding Award

Binghamton University

- Awarded summer internship funding to support a quantitative research internship with the First-year Research Immersion (FRI) program.

FEB 2026

Pi Mu Epsilon

National Mathematics Honor Society

FALL 2023 - PRESENT

Harpur College Dean's List

Binghamton University

- Named to the Dean's List every semester attended.

EACH SEMESTER

Provost's Promise Scholarship

Binghamton University

- Recipient of renewable financial support each semester throughout undergraduate study.

Leadership & Activities

SEP 2026 - PRESENT

Graphic Designer

Binghamton AI Society, Binghamton University

- Design promotional graphics and event materials, including posters, to support club events, outreach, and student engagement.
- Develop logo and branding concepts in collaboration with the executive board.

Technical Skills

Programming: R (tidyverse, tidymodels, caret), Python (Pandas, NumPy, scikit-learn, PyTorch), SPSS
Data Visualization & Applications: ggplot2, Plotly, Matplotlib, Observable JS, R Shiny, Qualtrics
Research & Reproducibility: Quarto, GitHub, LaTeX
Methods: Regression, random forests, decision trees, MARS, deep learning, differential expression, GSEA, mixed-methods analysis
Communication & Collaboration: Public speaking; Team-based research in academic and internship settings